System Architecture Specification

End-to-End Enterprise Technical Topology & Infrastructure Tiering

1. Architectural Principles & Mission

AuditEQ is built as an enterprise-grade, deterministic SEBI compliance auditing platform designed to audit 100% of equity trade pre-order phone calls. It decouples high-volume audio ingestion from real-time scoring, ensuring zero database deadlocks, cryptographic tamper resistance, and ultra-high phonetic tolerance across multilingual Indian telephony dialogue.

2. 4-Tier Enterprise Layer Topology

TIER 1: CLIENT

React 19 Enterprise Web Interface & HUD

Tailwind CSS v4 | Motion v12 | Lucide-React | HTML5 WebAudio Engine | Live SEBI Radar HUD

- Interactive Calls Player (Dual-Waveform, Stereo)
- Official SEBI 5-Pt Scorecards View
- Audited Master & Multi-Trade Grid
- Executive Analytics & Pareto Pareto Charts

↓ HTTPS / REST & WebSocket Streams (JSON + Audio Blobs)

TIER 2: GATEWAY

Express 4.21 High-Concurrency Node.js Application Server

TypeScript Execution | JWT Token Auth | Multer Stream Buffer | Rate Limiter | CORS & Headers

- Call & Audio Ingestion Controller
- Trade Reconciliation & Matching Router
- SEBI Audit & Scorecard Dispatcher
- PDF / Excel / Word Document Exporters

↓ In-Memory Pipeline Dispatch / Non-Blocking Worker Loop

TIER 3: PIPELINE

9-Stage Autonomous Deterministic SEBI Audit Pipeline

Dual-Engine ASR (Whisper + Gemini) | Speaker Diarization | Critical Entity Resolver | SEBI 5-Pt Scorer

- Dual-Channel PCM Audio Normalizer
- Groq Whisper Nova-3 Bilingual ASR
- Trade Timestamp & Multi-Clusterer
- Authoritative SEBI 5-Pt Scoring Engine

↓ PRAGMA WAL Direct Memory Map / SHA-256 Tamper Evident Writes

TIER 4: STORAGE

High-Concurrency SQLite WAL Engine & Immutable Audit Ledger

SQLite WAL Mode (60s busy_timeout) | Encrypted Audio Blob Vault | Tata Tele Webhooks | SMTP Relay

- Auditeq Production DB (WAL Mode)
- Local Storage Vault (.data/recordings)
- Tata Tele Smartflo CDR Webhooks
- Authoritative FundsIndia Directory

CONCURRENCY LOCK	AUDIT THROUGHPUT	ACTIVE DATA BASE	CARDINAL PARITY
0 Deadlocks	< 0.8ms	517 Calls	1 : 1 Parity
PRAGMA WAL + 60s timeout	Per 5-point evaluation	100% Speech Audited	Exactly 1 card per call

9-Stage Compliance Pipeline Architecture

Deterministic Processing from Raw Ingestion to Regulatory Scorecard

Pipeline Invariant Guarantee

Every physical call session transitions through 9 sequential, deterministic stages. A failure at any gate halts processing or routes the call to dedicated review, guaranteeing that non-pre-order calls never generate ungrounded scorecards and compliant calls receive mathematical scoring parity.

STAGE 1 — Ingestion & CDR Normalization

INPUT: Tata Tele / Smartflo CDR CSV or Webhook, Recording URL

OUTPUT: calls table record, duration, timestamps, calling_number

REGULATORY MANDATE:
Filter duration < 6s as SCRAP; assign initial status: IDLE.

STAGE 2 — Identity & Line Resolution

INPUT: Calling CLI (10-digit), Customer Phone, Trade Sheet

OUTPUT: client_code, registered_number, isRegisteredMatch

REGULATORY MANDATE:
Deterministic phone normalization (remove country codes, leading 0s).

STAGE 3 — Audio Conditioning & ASR

INPUT: Dual-Channel WAV/MP3, Groq Whisper Nova-3, Gemini Flash

OUTPUT: Full text transcript, word timestamps, confidence scores

REGULATORY MANDATE:
Speech normalization (Hinglish, financial vocabulary, numeral words).

STAGE 3.5 — Speaker Diarization & Attribution

INPUT: Left/Right Stereo Audio channels, Transcript word stream

OUTPUT: call_segments (ADVISOR vs CLIENT separated blocks)

REGULATORY MANDATE:
Strict speaker attribution: dealer statements separated from customer.

STAGE 4 — AI Call Intent Classification

INPUT: Transcript text, Dialogue structure, Call duration

OUTPUT: classification: PRE_ORDER | REGULAR | SCRAP

REGULATORY MANDATE:
Only verified PRE_ORDER calls advance to trade matching & compliance.

STAGE 5 — Trade Clustering & Reconciliation

INPUT: Trade executions table (time, stock, qty, price, dealer)

OUTPUT: matched_call_id, consolidated order execution clusters

REGULATORY MANDATE:
Multi-trade grouping: consolidate multi-executions into single session.

STAGE 6 — Audit Eligibility Gate

INPUT: Call transcript validation, duration >= 15s, trade match

OUTPUT: isAuditEligible: TRUE | BLOCKED

REGULATORY MANDATE:
Gated check: Missing audio or silent transcripts routed to Review.

STAGE 7 — Deterministic Q1–Q5 Compliance Audit

INPUT: Call dialogue segments, matched trade parameters, expected UCC

OUTPUT: q1..q5 statuses, evidence strings, confidence scores

REGULATORY MANDATE:
Evaluate SEBI regulatory questions with phonetic Levenshtein tolerance.

STAGE 8 & 9 — Authoritative Scoring & Publication

INPUT: Q1..Q5 results, SEBI fatal formula, dealer lookup

OUTPUT: scorecards row, audits row, PDF / XLSX report exports

REGULATORY MANDATE:
Strict cardinal constraint: Exactly 1 scorecard per physical call session.

Subsystems Deep-Dive: ASR & Consolidation

Critical Entity Resolution, Hindi Speech Normalization & Multi-Trade Parity

1. Authoritative Critical Entity Resolver (UCC & Trades)

Due to noisy 8kHz telephony recordings, verbatim regex matching causes high false failure rates. AuditEQ deploys a multi-stage acoustic normalizer that reconciles Indian financial speech artifacts:

Phonetic Homophone Mapping

Detects acoustic letter substitutions common in Indian telephony:

- "WIU", "VIA", "WAU", "WAD" -> Resolves to "WIA"
- "VIF" -> "WIF" | "VIC", "VIG", "WTG" -> "WIG"
- "PVA", "BWA", "PW8", "3W0" -> "PWA"

Speech De-Mangling Engine

Reverses automated ASR entity formatting:

- Currency symbol conversion: " ¹5,357" -> digits "5357"
- Telephony pauses: "WIA 6 0 7 2 2" stitched into "WIA60722"
- Removes commas, spaces & words like "Double a 82.04"

Levenshtein Acoustic Tolerance

Authoritative Levenshtein distance <= 1 matching on numeric and alphanumeric tokens:

- Catches 1-digit acoustic drops (e.g. PWA00930 vs PWA930)
- Confirms numeric ID accompanied by "client code / account" context.

Bilingual Order Intent Detection

Understands conversational trade execution instructions in Hinglish & Hindi:

- English: "buy", "sell", "CMP", "market price", "lots", "shares"
- Hindi: "' @‘R 9”,"Â) “ &” “ù ", "") “â &”²"Â)!"é2 ‘iK", "'©2'‰G ‘iK"

2. Multi-Trade Call Session Consolidation Architecture

In real-world equity brokerage, an advisor frequently receives trade orders for multiple stocks on a single phone call (e.g. Call #2 executing 3 trades, Call #28 executing 2 trades). The platform enforces a strict 1:1 cardinal constraint between physical calls and scorecards.

LEGACY BUG: 1 CARD PER TRADE

Trade-Keyed Scorecard Duplication

- Call #2 placed by client for 3 stocks.
- System created 3 duplicate scorecard rows with identical call_id.
- Client master export emitted duplicate caller names.
- Inflated total scorecard count and distorted audit pass percentages.
- Regulator received 3 separate audit certs for 1 physical recording.

PRODUCTION ARCHITECTURE: 1 CARD PER CALL

Consolidated Call Session Architecture

- Unique Constraint: uq_scorecards_call_id ON scorecards(call_id).
- Exactly 1 scorecard per physical call session.
- All trades executed during session consolidated into primary cluster.
- Symbols aggregated: "INFY / RELIANCE / TATASTEEL".
- Quantity summed, weighted average execution price computed.
- Single audit certificate signed with all executed order IDs.

CONSOLIDATION IMPACT: Reduced 1,173 redundant trade rows to 517 verified physical call sessions (0 duplicate calls, 100% data integrity).

Database Schema & Cryptographic Security

Entity-Relationship Topology, SQLite WAL Optimization & Audit Trail Immutability

1. Relational Database Schema & Cross-Table Keying

AuditEQ operates on an optimized relational schema in SQLite WAL mode. All compliance audits and scorecards are linked via authoritative foreign keys with strict cascade and unique indexes:

calls (Physical Audio Sessions)

id (PK) | call_date | call_time | phone_number | calling_number | registered_number | duration_seconds | transcript | classification (PRE_ORDER/REGULAR/SCRAP) | status | storage_path

trades (Exchange Order Ledger)

id (PK) | trade_date | trade_time | order_timestamp | client (UCC) | symbol | quantity | price | order_type (BUY/SELL) | dealer | advisor_name | matched_call_id (FK)

scorecards (Official 5-Point Audits)

id (PK) | call_id (UNIQUE FK) | audit_id (FK) | resolved_trade_id (FK) | caller_name | dealer | client_code | score (0..5) | is_fatal (0/1) | q1_status..q5_status | q1_evidence..q5_evidence

audits (Deterministic Regulatory Log)

id (PK) | call_id (FK) | trade_id (FK) | q1, q2, q3, q4, q5 (PASS/FAIL) | q1_flag..q5_flag | score (0..5) | compliance_disposition | status | model | sha256_hash

clients & advisors (Authoritative Master)

clients: id | client_code (UCC) | client_name | registered_mobile | advisor_name | dealer_id
advisors: id | dealer_code | full_name | email | team | phone

2. High-Concurrency SQLite WAL Engine Specification

To eliminate "SQLITE_BUSY: database is locked" errors under simultaneous audio transcription, live dashboard requests, and batch scorecard generation, AuditEQ configures mission-critical PRAGMA parameters:

`PRAGMA journal_mode = WAL;`
Enables Write-Ahead Logging; readers never block writers and writers never block readers.

`PRAGMA busy_timeout = 60000;`
Threads automatically wait up to 60 seconds to acquire write lock before timing out.

`PRAGMA synchronous = NORMAL;`
Reduces disk sync frequency while preserving database ACID crash durability.

`PRAGMA cache_size = -64000;`
Allocates 64MB dedicated RAM memory page cache for sub-millisecond query execution.

3. Cryptographic Tamper-Proofing & SEBI Audit Trail

SEBI inspection norms require brokerages to demonstrate that compliance audit scorecards and audio recordings have not been altered post-trade. AuditEQ embeds SHA-256 digital signing at every lifecycle transition:

- Raw Audio Hashing: SHA-256 fingerprint computed upon initial ingest; verified prior to playback.
- Immutable Scorecards: Completed scorecards cannot be altered without cryptographic revision logging.
- Export Signing: All PDF, Word, and Excel audit certificates carry cryptographic verification metadata.
- Audit Trail Logs: All operator actions, manual reviews, and email dispatches recorded in append-only table.